

Photoshop worklist

54 pages measured · ordered easiest fix first · 11 August 2026

Every correction I have measured, with the file, the pixel to erase, the pixel to draw to, and what is under each. Grouped by **how hard the Photoshop job is**, not by how bad the error is — so the top of this document is where to start.

Read the two shortcuts in section 2 before you open anything. On about a third of these pages the correct fix is to move the *label text*, not the leader line — which is a clean cut-and-paste on flat background instead of repainting over illustration. That converts the hardest class of defect into the easiest.

1 Before you open anything

Decide who owns each page — you or production

These are **flat JPEGs**. Production's fix is to re-render the page from their source scene, which keeps every other pixel intact. Your fix is to repaint the shipped JPEG. **If production later re-renders a page you have edited, your work is silently overwritten.** So pick a lane per page and tell them: either "we've fixed these ourselves, skip them" or "leave these to you". Doing both is the one outcome that wastes the effort.

My suggestion: **take groups A and B yourself** (short, self-contained, and they are the ones production is most likely to get wrong a second time), and **leave group C and everything in group D to them.**

Where the files are

All paths are relative to the root of the **private app repo** (rounds-codex-app). Most galleries live under assets/<id>/<id>-02.jpg; three of the pages here sit at the repo root instead, and those three are also the ones that ship undersized:

GALLERY	FILE	PAGE SIZE
most galleries	assets/<id>/<id>-02.jpg	1024 × 1536
tb	tb-02.jpg	800 × 1200
pneumothorax	pneumothorax-02.jpg	800 × 1200
hypothyroid	hypothyroid-02.jpg	800 × 1200

Every coordinate in this document is in the page's own pixels, origin top-left, which is what Photoshop's Info panel shows. For the three 800×1200 pages, multiply by 1.28 if you are working from a re-exported 1024×1536 file.

Three things to do after every edit

#	WHAT	WHY
1	Save as JPEG, quality 10-12 , same filename, same dimensions	The app and the service worker key off the path. Do not rename, do not resize.
2	Regenerate the thumbnail — gthumbs/<id>-02.jpg, 320 px wide, quality 82	The gallery grid reads the thumb, not the page. An unregenerated thumb shows the old artwork on the card.
3	Rebuild the gallery PDF — scripts/rebuild_gallery_pdf.py <root> <id>	This is the one that has already bitten us. The download PDF is a build artifact generated once. Four galleries shipped for months with the old logo in the PDF and the new one in the viewer, because nothing regenerated the PDF. Send me the edited pages and I will run steps 2 and 3.

Then python3 scripts/verify_gallery_pdfs.py <root> checks every page of every gallery PDF against the page the app serves — it will catch it if step 3 was missed.

2 Two shortcuts that save most of the work

Shortcut 1 — swap the label text, not the leader

Where two labels have landed on *each other's* structure, both leaders are already correct in the sense that each ends on a real structure that the *other* label names. So the fix is to **exchange the two pieces of label text** and leave every line untouched.

Label text sits on flat dark panel background. Lasso, cut, paste, nudge into place, heal the two vacated rectangles. **No painting over illustration at all.**

PAGE	SWAP THESE TWO LABELS	NOTE
cellulitis p2	Artery ↔ Vein	Single words, stacked one above the other in the right-hand column. The cleanest one on this list — do it first.
ckd p2	Bowman's capsule ↔ Glomerulus	RENAL CORPUSCLE panel. A pure swap.
cdiff p2	Normal mucus layer ↔ Intact epithelium	NORMAL COLONIC MUCOSA panel. A pure swap.
osteoporosis p2	Tibia ↔ Fibula	Single words, stacked. But see group 5 — confirm which bone is which on the lateral skeleton first, and note that on the <i>anterior</i> skeleton both dots are on one bone, so a text swap alone will not finish that half.

Shortcut 2 — shift a whole label column by one position

Three pages have a column of labels where **every label has taken the next one's structure**. Rather than moving several leaders, move the text down (or up) one slot in the column and fix only the one label at the end of the chain that has nowhere to go.

PAGE	THE CHAIN	WHAT IS LEFT OVER
labor p2	Sacrum → a lumbar vertebra · Rectum → the sacrum · Vagina → the rectum	Shift the three right-hand labels up one slot, then give the vagina its own leader. Erase the topmost dot at (732,341).
withdrawal p2	HYPOTHALAMUS → pink · AMYGDALA → green · LOCUS COERULEUS → orange · CEREBELLUM → orange · BRAINSTEM → nothing	Shifting the text fixes three of five. Two leaders still need moving: one onto the blue blob at (437,495) and one onto the mauve cerebellum at (505,530) , neither of which currently has any leader.
cdiff p2	Sigmoid colon → the descending limb · Rectum → haustrated bowel · Anal canal → sigmoid	Shift the three down one slot. Descending colon at (462,348) is correct and stays.

One caution on both shortcuts. Check the label's own ink colour before you move it. On several of these pages the label is printed in the colour of the structure it names — *withdrawal* p2 is entirely built that way — so moving the text without recolouring it would leave the colour wrong even though the words are right.

3 The erase-and-repaint method

Where you do have to move a line, this is what I learned doing about a dozen of these with scripts. Photoshop's healing brush will do most of it better than my code did, but the choice of *donor* is the whole job.

WHAT IS UNDER THE LINE	HOW TO ERASE IT
Flat black background	Anything works. Spot healing brush, one pass.
Flat or smoothly shaded tissue	Healing brush with a donor 20–30 px away, perpendicular to the stroke.
Textured tissue (gyri, trabeculae, folia)	Clone stamp, donor parallel to the stroke so the texture direction survives. Sample often; a single donor smears.

WHAT IS UNDER THE LINE	HOW TO ERASE IT
Banded or striated tissue (adrenal cortex, muscle, arbor vitae)	Clone along the band , not across it. Find the grain by eye first — on <i>addisons</i> p2 the leaders ran <i>parallel</i> to the striations, so the donor had to be perpendicular. The rule is "clone along the texture's grain", and the grain sometimes runs with the stroke.
A strong edge crosses the stroke — an organ boundary, a vessel wall, a bone cortex	Do not erase it. Any donor reconstructs one side and destroys the other. I abandoned two fixes for exactly this. Leave those for a re-render.

Erase the whole polyline, not the straight line between its ends. Several of these leaders bend. On *pancreatitis* p2 I masked a straight line between the endpoints and left a white diagonal across the aorta — the bounding box and the pixel count both looked right, and only a side-by-side caught it.

Check your work at 100%, then at 25%. A repaint that is invisible at 100% often shows as a smudge at thumbnail size, which is where most readers see the page first.

4 Group A — move one dot, easy background, short move

Start here. Each is one terminator, on flat or dark ground, moving under about 40 px. Fifteen minutes each.

PAGE	LABEL	ERASE THE DOT AT	WHAT IS THERE NOW	DRAW IT AT	WHAT IS THERE
seizure p2	Axon	(253,1143)	a tan myelin internode	(268,1152)	a node of Ranvier
seizure p2	Neurotransmitters	(527,1080)	the bouton rim, nothing there	(495,1089)	a vesicle circle
seizure p2	Synaptic cleft	(530,1131)	12 px short, dark ground	(500,1130)	the orange particle field
depression p2	Nucleus accumbens	(493,424)	the dark gap below the sphere	(492,401)	the violet sphere
depression p2	Hippocampus (coronal)	(905,415)	dark cortex/white matter	(884,434)	the bright purple hippocampus
stroke p1	Thrombus blocking the MCA	(556,368)	the dark red infarct core	(548,375)	the artery, 8 px away
stroke p1	Ischemic core	(893,631)	the plain pink left hemisphere	(955,617)	the dark red core
stroke p1	Penumbra	(890,730)	the plain pink left hemisphere	(975,720)	the gold rim
migraine p2	C2 (cervical panel)	(150,1264)	red neck muscle	(150,1280)	the yellow C2 ramus. C3 just below shows the intended form
migraine p2	F — Trigeminal ganglion	(453,589)	the beige nerve trunk	(448,548)	the ringed ganglion hub
migraine p2	C — Middle meningeal artery	(598,412)	bare cortex	(588,428)	the red arterial tree
osteomyelitis p2	Endosteum	(343,533)	inside the gold marrow block	(359,533)	the inner surface of the cortical shell
osteomyelitis p2	Volkman's canal	(783,622)	a black osteocyte lacuna	(766,548)	the red/blue vessel pair
nephrolithiasis p2	Medulla (renal pyramids)	(818,856)	the cortical band	(795,860)	inside a pyramid
nephrolithiasis p2	Urinary bladder	(278,871)	the tan distal ureter	(320,890)	inside the red bladder
uti p2	③ Renal vein	(413,413)	the red artery's lower edge	(430,426)	the blue vein, 12 px below
uti p2	Renal pelvis	(844,347)	a red pyramid	(793,350)	the tan funnel
uti p2	Major calyx	(842,390)	a red pyramid	(803,372)	a tan branch
uti p2	Minor calyx	(830,417)	a red pyramid base	(816,378)	a tan cup
hepatitis p2	Sinusoids	(413,941)	inside a salmon plate block	(403,948)	the blue radial channel

PAGE	LABEL	ERASE THE DOT AT	WHAT IS THERE NOW	DRAW IT AT	WHAT IS THERE
hepatitis p2	Hepatocytes (plates)	(424,888)	a blue portal venule	(405,915)	a salmon plate block
hepatitis p2	Left hepatic duct	(105,883)	the common hepatic trunk	(85,866)	the left-going limb
cdiff p2	Taenia coli	(169,366)	the haustral wall	(189,366)	the tan longitudinal band, 19 px medial
cdiff p2	Ileocolic a.	(139,1143)	black background	(152,1152)	the arcade branch — easiest erase on the page
icp p2	CSF ~10%	(770,468)	pink parenchyma	(795,469)	the blue CSF band
icp p2	BRAIN ~80%	(783,369)	the blue CSF band	(740,400)	the pink parenchyma
hypothyroid p2	Parafollicular (C) cell	(126,672)	the pale colloid	(160,710)	the green C cell
parkinsons p2	Subthalamic nucleus	(347,540)	a tan wedge	(317,566)	the green sphere matching the label's ink
suicide p2	ACC	(339,455)	the cingulate band's midpoint	(262,470)	the rostral limb of the same band
suicide p2	OFC	(200,507)	mid medial-frontal cortex	(210,545)	the orbital surface
gdm p2	② UMBILICAL CORD	(373,762)	the fetal back	(435,700)	the purple coiled cord
preeclampsia p2	UMBILICAL CORD	(285,879)	maternal pelvic vessels	(372,852)	the cord at the fetal abdomen
appendicitis p2	Submucosa	(185,900)	the orange lumen (radius 51)	radius 76–90	the dark red band. Circle centre (134,900), radius 110

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Group B — move one dot, textured background

Same job, but the old dot sits on gyri, folia, trabeculae or bowel wall. Clone stamp with the grain; see section 3.

PAGE	LABEL	ERASE AT	WHAT IS THERE NOW	DRAW AT	WHAT IS THERE
dementia p2	Corpus callosum	(292,424)	the purple cingulate arc	(250,442)	the salmon callosal band
dementia p2	Cingulate gyrus	(261,478)	the blue thalamus	(250,415)	the purple arc
dementia p2	Thalamus	(313,478)	the salmon corpus callosum	(270,478)	the blue oval
dementia p2	Brainstem	(312,590)	a purple blob	(290,600)	the gold column
dementia p2	Hippocampus	two rings, (376,566) and (388,573)	straddling green rim and pink cerebellum	one ring at (330,545)	inside the green body
dementia p2	Lateral ventricles	(730,472)	a purple hippocampus	(707,425)	a dark ventricular cavity
dementia p2	Hippocampus (bilateral)	(805,458)	the midline	(735,468) and (880,468)	the two purple bodies
dementia p2	Entorhinal cortex (bilateral)	(883,470)	a purple hippocampus	(762,468) and (855,468)	the gold bands
schizophrenia p2	Corpus Callosum	(489,424)	the blue thalamic mass	(489,386)	the ribbed callosal band, straight up the same leader
schizophrenia p2	Thalamus	(618,428)	parietal cortex	(495,443)	the blue thalamic mass
schizophrenia p2	Cerebellum	(583,684)	the brainstem/cord tube	(640,530)	the cerebellar folia
parkinsons p2	Globus pallidus (external)	(226,493)	pink frontal cortex	(283,545)	the purple lump chain
parkinsons p2	Thalamus (left column)	(300,548)	the pallidum	(368,528)	the blue body (the key's thalamus colour)

PAGE	LABEL	ERASE AT	WHAT IS THERE NOW	DRAW AT	WHAT IS THERE
parkinsons p2	Thalamus (right column)	(396,551)	the red nucleus's edge	(368,528)	the blue body
parkinsons p2	Cingulate gyrus	(290,420)	the purple callosal band	~15 px up and out	the pink cortical ribbon outside the arc
depression p2	DLPFC	(473,323)	the green cingulate band	the blue frontal convexity	above and in front of the band
depression p2	vmPFC	(577,323)	the band's posterior end	≈(465,375)	ventral-anterior frontal cortex
depression p2	Anterior cingulate cortex	(461,371)	blue frontal cortex	(473,323)	the green band — where DLPFC currently ends
suicide p2	Hippocampus	(359,578)	the violet raphe sphere	(305,580)	the plum medial-temporal band
suicide p2	Amygdala	(245,571)	empty dark background	(305,556)	the orange sphere
suicide p2	DLPFC	(402,472)	the posterior cingulate band	(240,435) or (215,470)	rose convexity or blue frontal lobe — your call
osteomyelitis p2	Yellow marrow	(353,441)	deep red marrow	(328,505)	the gold block (x 313–347, y 478–560)
osteomyelitis p2	Haversian canal	(738,421)	the lamellar rings	(790,402)	inside the dark canal opening
osteomyelitis p2	Venous drainage	(285,1250)	distal cortical shell	(386,1080) or (344,1180)	the medullary venous channels
hepatitis p2	Cystic duct	(265,523)	inside the gallbladder sac	(325,496)	the tube at the neck
hepatitis p2	Common bile duct	(349,524)	above the confluence	(366,555)	below the cystic-duct junction
hepatitis p2	Right hepatic artery	(355,537)	the green bile duct	clear of the duct	the dark maroon arterial twig ~5 px up-right
hepatitis p2	Bile duct (L and R hepatic)	(408,535)	the blue IVC	(366,545)	the green trunk, 42 px left
hepatitis p2	Inferior vena cava	(376,567)	the orange triad widget	(411,567)	the blue IVC trunk
hepatitis p2	Hepatic portal triad	(390,537)	the red artery alone	(376,567)	the triad widget the IVC label vacates
ckd p2	Proximal tubule	(821,431)	a blue vessel	(858,431)	the orange-walled tubule
ckd p2	RENAL VEIN	(275,510)	kidney parenchyma	(365,500)	the blue renal vein
ckd p2	RENAL ARTERY	(292,437)	a medullary pyramid	(324,429)	the arterial trunk at the hilum
nephrolithiasis p2	Renal artery	(198,489)	kidney parenchyma	(295,458)	the red renal artery
nephrolithiasis p2	Renal vein	(325,572)	the vertical IVC	(300,470)	the horizontal renal vein
hyperkalemia p2	② Renal Medulla	(347,451)	the intact kidney's capsule	(533,435)	a pyramid in the cut kidney
hyperkalemia p2	④ Collecting Duct	(908,632)	the DCT coil	(960,620)	the branched collecting duct
gerd p2	② Diaphragmatic crus	(444,319)	the oesophageal wall	(395,365)	the crural band
gerd p2	④ Lower esophageal sphincter	(497,377)	shared with ⑤ and ⑥	(470,340)	the thickened distal segment
gerd p2	Widened hiatal opening	(240,1032)	the gastric body	(195,1000)	the gap between the crura
gerd p2	Diaphragm (overview)	(105,437)	the stomach body	(100,417)	the dome — confirm one is drawn
iron-anemia p2	JEJUNUM	two dots, (196,670) and (204,652)	haustrated colon	one dot, ≈(300,720)	the smooth coiled loops
preeclampsia p2	HEART	(322,529)	the right lung	(390,580)	the ventricular mass

PAGE	LABEL	ERASE AT	WHAT IS THERE NOW	DRAW AT	WHAT IS THERE
preeclampsia p2	AORTA	(317,597)	the right lung	(372,528)	the arch
preeclampsia p2	PELVIC CIRCULATION	(452,841)	the uterine wall / fetal buttock	(320,930)	the iliac network
tb p2	Alveolar macrophage	(403,674)	a pink alveolus	(368,678)	the purple macrophage. 800×1200 page
pneumothorax p2	Cervical recess	(608,878)	shoulder soft tissue	(565,877)	the blue apical patch. 800×1200 page
pneumothorax p2	Costophrenic recess	(597,963)	lung above the crescent	(597,985)	the blue crescent
pneumothorax p2	Costomediastinal recess	(597,925)	mid-zone lung	(562,925)	the reflection beside the mediastinum
pneumothorax p2	Costophrenic angle	(682,402)	the lateral lung border	(674,438)	the angle itself
pneumothorax p2	Intercostal vein	(283,873)	the subcutaneous band	(270,906)	the blue segment in the groove
pneumothorax p2	Rib (bundle panel)	(288,963)	magenta lung	(275,915)	the gold rib band
labor p2	Bladder	(475,559)	the cream pubic bone	the bladder lumen	confirm one is drawn

6 Group C — longer re-routes

The leader travels far enough that you are redrawing a stroke, not just a dot. Doable, but slower, and the erase crosses more artwork. Consider handing these to production.

PAGE	LABEL	FROM	TO	WHY IT IS HARDER
parkinsons p2	Brainstem	(245,726)	(358,600)	170 px. Old dot is on black — erase is trivial — but the whole leader must be redrawn.
parkinsons p2	Cerebellum	(303,760)	(455,595)	225 px, same shape.
stroke p1	Middle Cerebral Artery	(447,423)	(520,410)	Also recolour the terminator white — it is currently gold, the same amber the figure uses for the penumbra, which is why it reads as artwork.
cdiff p2	Superior mesenteric a.	(122,1173)	(192,1150)	70 px across the arcade.
cdiff p2	Superior mesenteric v.	(478,1107)	(544,1120)	66 px.
cdiff p2	Inferior mesenteric v.	two dots, (481,1143) and (481,1160)	(510,1202)	Two terminators to erase; one to draw.
cdiff p2	Parasympathetic (vagus n.)	(851,1110)	(690,1140)	160 px across the whole colon frame.
cdiff p2	Portal vein (to liver)	(531,1202)	(545,1088)	Erase is easy — black ground — but the new route runs up the panel.
cdiff p2	Right colic a.	two tips, (122,1117) and (142,1118)	one tip on an arcade branch	Remove the orphan dash.
ckd p2	Afferent / Efferent arteriole	(865,1024) and (872,1063)	(775,970) and (770,1012)	Both leaders cross the whole corpuscle to reach the vascular pole on the far side.
seizure p2	Cortical networks	starts (531,891), arrowhead at (583,891) pointing outward	(555,845), arrowhead inward	Reverse the arrow. Not a move — the head is at the wrong end.
seizure p2	Basal ganglia	(496,452)	(552,452) or (545,478)	Crosses the magenta thalamus.

PAGE	LABEL	FROM	TO	WHY IT IS HARDER
migraine p2	A — Dura mater	(521.5,306.5)	(470,283)	The dural band is only 6 px thick (y 280–285 at x 470). Precision job.
suicide p2	Raphe / others	—	—	See the per-page sheet; several leaders share one region.
hypothyroid p2	Capillary	(170,712)	(60,715)	110 px across the follicle. 800×1200 page.

7

Group D — do not attempt in Photoshop

The label names something the page does not draw

Nine rows. There is nothing to point at, so no leader move fixes them. These need an illustrator or a decision to drop the label. Full detail in `PRODUCTION-artwork-needed.md`.

PAGE	LABEL	WHAT IS MISSING	SUGGESTED
stroke p1	Circle of Willis, Basilar, PCA, ACA	all four — a lateral view cannot show basal vessels	add a basal inset, or drop the four
uti p2	® Ureterovesical junction	no oblique intramural segment	draw it — the page's own relationships strip shows how
parkinsons p2	Substantia nigra pars reticulata	no SN band in the sagittal panel	copy from the page's own midbrain panel
parkinsons p2	Internal capsule	no white-matter lamina	draw one, or move the label to the coronal panel
schizophrenia p2	Pituitary Gland	no gland or stalk	draw it; the connector arrow is fine as is
migraine p2	D — Superficial temporal artery	no extracranial vessel anywhere	draw the STA in the scalp, or drop the label
hepatitis p2	Coronary ligament	a posterior structure in an anterior view	drop the label — it cannot be drawn here
hypothyroid p2	Hyoid bone	no bone above the thyroid cartilage	drop the label
dementia p2	Fornix	not differentiated from the callosal band	draw it, or fold into "Corpus callosum (fornix beneath)"

A legend entry whose marker was never drawn

Five. You *could* add a numbered badge in Photoshop — it is a circle and a digit on flat artwork — but check first whether the number should exist at all.

PAGE	MISSING MARKER
gerd p2	⑦ Stomach (cardia) — the artwork has ① to ⑥ only
meningitis p2	④ Cerebral aqueduct — no ④ anywhere
meningitis p2	⑥ Median aperture — no ⑥ anywhere
lymphoma p2	⑤ Mediastinal Lymph Nodes — the column runs 2,3,4,6,7,8,9
gout p2	⑤ Other metatarsals — no ⑤ badge on the foot

Blocked on a re-export, not on Photoshop

pneumothorax p2, rows 5–8. The four pleural layers on the main figure are **four pale lines one or two pixels wide with three-pixel gaps** — a terminator on that page is 4 px across, wider than the layer it names. They cannot be verified or drawn unambiguously at 800×1200. Re-export at 1024×1536 first. Same cause blocks four rows on *hypothyroid* p2 and one on *tb* p2.

Waiting on your read, not on anyone's tools

gdm p2, the PLACENTA (ENLARGED) panel. Either the umbilical vessel colours are inverted or the two leaders are on the wrong vessels — opposite fixes, and a render cannot distinguish them. Remember the umbilical circulation reverses the usual convention: the **vein** carries oxygenated blood and is drawn **red**.

The placenta on both obstetric pages. On *preeclampsia* p2 and *gdm* p2 the PLACENTA label lands on the uterine wall and I could not find a distinct placental disc on either. If it is not drawn, both are artwork rows.

osteoporosis p2, the lateral skeleton. Both legs overlap and I could not be certain which shaft belongs to the near leg. Confirm which bone is which before swapping Tibia and Fibula, and note that on the *anterior* skeleton both dots are on the slender bone — so that half needs a leader move as well as a text swap.

Compiled from 47 per-page production sheets, 11 August 2026. Each page's full reasoning, colour samples and unverified rows are in galleries-staging/PRODUCTION-<id>-p2.md on the `claude/anatomy-label-corrections-j58lf1` branch. 54 of the 81 pages in the work order are measured; 27 are not yet, so this list will grow.